MPHOENTLE PALESA KEIKABILE

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PROFESSIONAL SUMMARY

Final-year Bachelor of Engineering Technology student in Computer Engineering at the Cape Peninsula University of Technology (CPUT), with hands-on exposure to embedded systems, software design, and network, electronic communications and data systems. Brings a strong foundation in leadership and teamwork, along with sound problem-solving skills and a keen interest in building practical, technology-driven solutions.

Additional academic experience in SQL database development, data reporting, Python-based network monitoring and technical research.

EDUCATION

Bachelor of Engineering Technology (BEngTech) - Computer Engineering
Cape Peninsula University of Technology (CPUT) | Final Year, Expected Completion 2026

National Senior Certificate (Matric)
Completed 2022

TECHNICAL SKILLS

- Programming Languages: C#, C++, Java, Python, JavaScript (WebGL), HTML, CSS
- SQL (Structured Query Language): database queries, joins, grouping, filtering, aggregates and common table expressions (CTEs)
- Relational Database Design, Data Retrieval & Reporting
- Embedded Systems Development
- Digital Systems & Electronics
- Data Systems
- Network Systems
- Software Design & Development
- Microsoft Word & Excel

CORE COMPETENCIES

- Team Leadership & Collaboration
- Problem-Solving & Analytical Thinking
- Time Management & Adaptability
- Communication
- Technical Research, Testing & Report Writing

TOOLS & PLATFORMS

- MySQL
- GNS3
- Cisco Packet Tracer
- Linux
- Tinkercad Software Simulation
- Multisim
- MySQL Workbench & Draw.io (diagrams.net)
- VyOS & OpenDaylight (software-defined networking)

ACHIEVEMENTS & LEADERSHIP (HIGH SCHOOL)

Member, Student Executive Committee *2020 – 2021*

- Represented fellow learners in school governance matters, contributing to decision-making and school initiatives.
- Developed early leadership, organisational, and communication skills through active committee participation.

PROFESSIONAL EXPERIENCE

Administrative Volunteer | High School Administration Office

- Assisted with capturing and updating learners' year marks accurately on the school's administrative system.
- Registered and updated learner information in the school's database for the new academic year.
- Compiled and organised high school email addresses and related administrative contact information.
- Provided general administrative support to the school office while maintaining accuracy and confidentiality of learner information.

PROJECTS

Arduino-based solar powered washing machine controller | Embedded Systems *2025*

- Designed and programmed an Arduino-based solar-powered washing machine controller (C/C++, Tinkercad simulation and physical prototype) that adapts its wash, rinse, and spin cycles to real-time solar availability received over infrared, pausing and resuming to minimise grid and battery use.

Interactive Snake Game | WebGL, JavaScript, HTML/CSS | Graphics Design *2025*

- Built an interactive Snake game using WebGL for GPU-accelerated 2D/3D graphics, with HTML and CSS for the page layout, UI, and styling.

AI – based Smart Home Automation System | Industrial Design *2026 – In Progress*

- Designed and built an AI-powered smart home automation system on ESP32 (C/C++, Python, MQTT) that uses sensor-driven behaviour learning and solar-aware scheduling to automatically control four appliances (solar geyser, chest freezer, washing machine, water pump) via a 4-channel relay module, incorporating PIR, DHT22, LDR, and ACS712 current sensors for occupancy, environmental, and fault-detection input, with the full circuit simulated and validated in Cirkit Designer prior to physical hardware assembly.

SDN-Driven Virtual Network | Network Systems 3 *2025*

- Built and tested a four-router virtual network in GNS3 using VyOS and IS-IS dynamic routing; connected the OpenDaylight SDN controller using BGP.
- Used Python to retrieve information from the network controller and investigated IP addressing and router-configuration problems.

Infrastructure Maintenance Database | Database Systems 3 *2025*

- Created a MySQL database for maintenance requests, locations, service teams and assets; used SQL queries to report issue frequencies, request statuses and overdue tasks.
- Compared MySQL Workbench and Draw.io for database management and visual modelling, and documented the design and results.

Online Store Database | Database Systems 3 *2025*

- Created MySQL tables for customers, products, orders and order items; used SQL common table expressions (CTEs) and aggregate queries to calculate customer spending.
- Implemented a stock-update stored procedure and a discount function, and explored database triggers and automated events.

CERTIFICATIONS

- Cisco Networking Foundations: IP Addressing | LinkedIn Learning | April 2025
- Learning IP Addressing | LinkedIn Learning | April 2025

INTERESTS & HOBBIES

- Exploring new programming languages and personal coding projects
- Tinkering with electronics and small embedded system builds
- Reading about emerging technology trends
- Puzzle-solving and strategy games